

A BAYESIAN APPROACH TO SHRINKAGE ESTIMATION OF THE SCALE PARAMETER OF A WEIBULL DISTRIBUTION

M. Pandey : Department of Zoology, Banaras Hindu University, Varanasi, India.

Key words : Exponential density; maximum likelihood estimator; Monte Carlo methods.

Summary : Sometimes a prior value θ_0 of a parameter θ is available. It is found that shrinkage estimators $\hat{\theta}_s = k\hat{\theta} + (1-k)\theta_0$ are more efficient than the unbiased estimator $\hat{\theta}$ around $\theta = \theta_0$. A value of k ($0 < k < 1$) is chosen depending on the faith one has in the value θ_0 . This paper considers a prior assumption that $(1-p)$ is the probability that the scale parameter Θ of a Weibull distribution (with known shape parameter), has the value θ_0 and the rest of the probability mass p ($0 < p < 1$) is distributed as an exponential density. With this prior density the Bayes estimator $\hat{\theta}_B$ is obtained first and the Bayesian shrinkage estimator of θ is then defined by replacing θ_0 in $\hat{\theta}_s$ by $\hat{\theta}_B$. The mean squared error comparison of the proposed estimator with the ordinary shrinkage and the ordinary Bayes estimators shows that it is better than the latter two if the shrinkage factor and the prior density parameters are properly chosen.

1. Introduction

If a prior value θ_0 of a parameter θ is known, Thompson (1968) suggested shrinking the minimum variance unbiased linear estimator of θ towards

θ_0 . The resulting shrinkage estimator $T_s = k\hat{\theta} + (1-k)\theta_0$, ($0 < k < 1$) is more efficient around $\theta = \theta_0$. Mehta and Srinivasan (1971) modified and generalised Thompson's estimator by introducing an exponential term in order to bound its relative efficiency (REF). Their estimator becomes a little more complicated for computation. Lemmer (1981) examined another modification of the shrinkage estimator suggested by Pandey (1979), $T_p = a[k\hat{\theta} + (1-k)\theta_0]$ and found that for the normal variance problem the choice of 'a' must be close to 1, which reduces T_p to T_s .

Prior knowledge about θ can be more realistically expressed by specifying a prior distribution for θ around θ_0 instead of specifying a point value θ_0 . For stressing one's belief in θ_0 , a mass $1-p$ may be put on the value θ_0 and the rest of the probability mass p distributed according to a probability density. Lemmer (1981) suggested replacing θ_0 in T_s by T_B , the Bayes estimator of θ with respect to this prior. He applied this Bayes' shrinkage procedure to estimate normal and Poisson means and the binomial proportion.

As the Weibull distribution has a wide applicability in life-testing problems, this paper considers a Bayes shrinkage estimator of its scale parameter when the shape parameter is known. It is defined in section 2. Section 3 deals with comparison of estimators. Five hundred random samples have been generated from a Weibull distribution by Monte Carlo methods and the REF of the Bayes shrinkage, ordinary shrinkage and ordinary Bayes estimators with respect to the maximum likelihood estimator (MLE), which is also unbiased, have been obtained for various values of the pertinent parameters. Numerical findings are summarized in section 4, and section 5 contains recommendations for the choice of the parameters.

2. Bayes shrinkage estimators

Let $x_1, x_2, \dots, x_n$ be a random sample from a Weibull distribution with p.d.f.

$$(2.1) \quad f(x/\beta, \theta) = \frac{\beta}{\theta} x^{\beta-1} \exp\left(-\frac{x^\beta}{\theta}\right); \quad \beta, \theta > 0, \quad x > 0.$$

The shape parameter β is assumed to be known.

Suppose the prior density of θ is given by

$$(2.2) \quad g(\theta) = \begin{cases} \frac{pq \exp(-a/\theta)}{\theta^c}; & c > 1, a, \theta > 0 \text{ if } \theta \neq \theta_0 \\ 1 - p & \text{if } \theta = \theta_0 \end{cases}$$

where $q = a^{c-1}/\Gamma(c-1)$.

The likelihood of the sample, $L(\underline{x}/\theta)$ is given by

$$(2.3) \quad L(\underline{x}/\theta) = \left(\frac{\beta}{\theta}\right)^n \prod_{i=1}^n x_i^{\beta-1} \exp\left(-\sum_{i=1}^n \frac{x_i^\beta}{\theta}\right)$$

Then for squared error loss the Bayes estimator of θ for the above prior is given by

$$(2.4) \quad E(\theta/x) = \frac{\int_0^\infty \theta L(\underline{x}/\theta) g(\theta) d\theta}{\int_0^\infty L(\underline{x}/\theta) g(\theta) d\theta}.$$

After simple algebra this gives the Bayes estimator as

$$(2.5) \quad \hat{\theta}_B = [qpAU + (1-p)\theta_0 WU^m] / [qp(m-1)A + (1-p)WU^m]$$

where

$$A = \theta_0^n \Gamma(n+c-2),$$

$$U = \left(\sum_{i=1}^n x_i^\beta + a \right),$$

$$W = \exp\left(-\sum_{i=1}^n \frac{x_i^\beta}{\theta_0}\right),$$

$$m = n + c - 1.$$

The MLE of θ , which is also unbiased is given by

$$(2.6) \quad T_{ML} = \sum_{i=1}^n x_i^\beta / n.$$

Hence the ordinary shrinkage estimator of θ is

$$(2.7) \quad T_s = k T_{ML} + (1-k)\theta_0; \quad 0 < k < 1.$$

Replacing θ_0 in (2.7) by $\hat{\theta}_B$ we define the Bayes shrinkage estimator of θ as

$$(2.8) \quad T_{BS} = k T_{ML} + (1-k)\hat{\theta}_B.$$

If $p = 0$, $T_{BS} = T_s$, and, if $p = 1$ and $k = 0$, T_{BS} reduces to the ordinary Bayes estimator of θ defined earlier by Sinha and Kale (1980) as

$$(2.9) \quad T_{OB} = \frac{n T_{ML} + a}{n + c - 2} = U/(m-1).$$

3. Comparison of estimators

For studying the efficiency of T_{BS} , the mean squared errors (MSE) of T_{BS} , T_s and T_{OB} are compared with that of T_{ML} , the REFs E_1 , E_2 and E_3 being defined as

$$(2.10) \quad E_1 = V(T_{ML})/MSE(T_s)$$

$$(2.11) \quad E_2 = V(T_{ML})/MSE(T_{OB})$$

$$(2.12) \quad E_3 = V(T_{ML})/MSE(T_{BS}).$$

It is not feasible to determine E_1 , E_2 , E_3 by analytical methods so

Monte Carlo methods have been used. These efficiencies are based on 500 simulated samples of sizes $n = 10, 20, 40$ from a Weibull distribution with $\beta = 2$. However, choice of β does not affect the efficiencies. θ_0 is chosen as 5,00 and efficiency calculations are done for $\theta/\theta_0 = 0,25(0,25) 2,0$; $k = 0,25, 0,50, 0,75$; $p = 0,25, 0,50, 0,75$; $a = 2(2)8$ and $c = 2, 3, 4$.

Complete results are too extensive to be published so for illustration only one table and some graphs are given for selected values of various parameters.

4. Numerical findings

Variation of efficiencies with respect to n, a, c, p, k was studied for values of θ/θ_0 and figures 1 through 5 depict it in the said order.

It is found that E_1 and E_3 are maximum at $\theta/\theta_0 = 1$, as expected. Figure 1 shows that T_{BS} is more efficient than T_s for $\theta/\theta_0 < 1,00$ for $n = 10$ and for $\theta/\theta_0 < 1$ and $\theta/\theta_0 > 1,25$ for $n = 20$ and 40. But with increasing sample size the length of effective interval of T_{BS} is more efficient than T_{OB} if $\theta/\theta_0 \approx 1,0$ and the interval of θ/θ_0 where $E_3 > E_2$ becomes shorter with larger values of n . Moreover, E_2 remains almost equal to 1 throughout the range of θ/θ_0 , whereas E_3 and E_1 are quite high if the prior value θ_0 is also the true value of θ .

Figure 2 shows that E_3 increases with increasing value of a if $\theta/\theta_0 < 1,0$ or $\theta/\theta_0 > 1,50$. The effective interval for T_{BS} is $0,75 \leq \theta/\theta_0 \leq 1,50$ for all a . T_s is more efficient than T_{BS} in this interval whereas outside this interval T_{BS} is more efficient for $a = 8$. Though T_{OB} has a longer effective interval its maximum efficiency is almost the same as that of MLE.

Figure 3 shows the variation of efficiencies for various choices of c which is similar to that of figure 2. With increasing value of c E_3 increases in the effective interval. $c > 2$ makes T_{OB} more efficient than T_{ML} but the gain is little compared to maximum gain in efficiencies of T_{BS} and T_S around $\theta/\theta_0 = 1$.

Figure 4 shows that with increasing value of p the gain in efficiency of T_{BS} decreases in the effective interval but outside this interval E_3 increases. At $\theta/\theta_0 \leq 0,50$, T_{BS} is more efficient than T_S for all p .

Variation in efficiencies with k is shown in figure 5. It is evident that $k = 0,25$ results in the highest efficiency of T_{BS} in the effective interval of θ/θ_0 , whereas $k = 0,75$ leads to highest efficiency of T_{BS} outside effective interval. If $k < 0,50$, T_{BS} is more efficient than T_S for $\theta/\theta_0 \leq 0,50$.

5. Recommendations

On the basis of these findings the following recommendations are made:

If it is expected that $0,75 \leq \theta/\theta_0 \leq 1,50$ choice of k and $p = 0,25$ will lead to large gain in efficiency of the proposed estimator over MLE . Small samples should be preferred because their use widens the effective interval of θ/θ_0 . The highest gain in efficiency of the proposed estimator is achieved if the prior density has small $a(=2)$ and large $c(=4)$.

If it is feared that the true value of θ may differ largely from the prior value θ_0 , k and p should be taken as $0,75$.

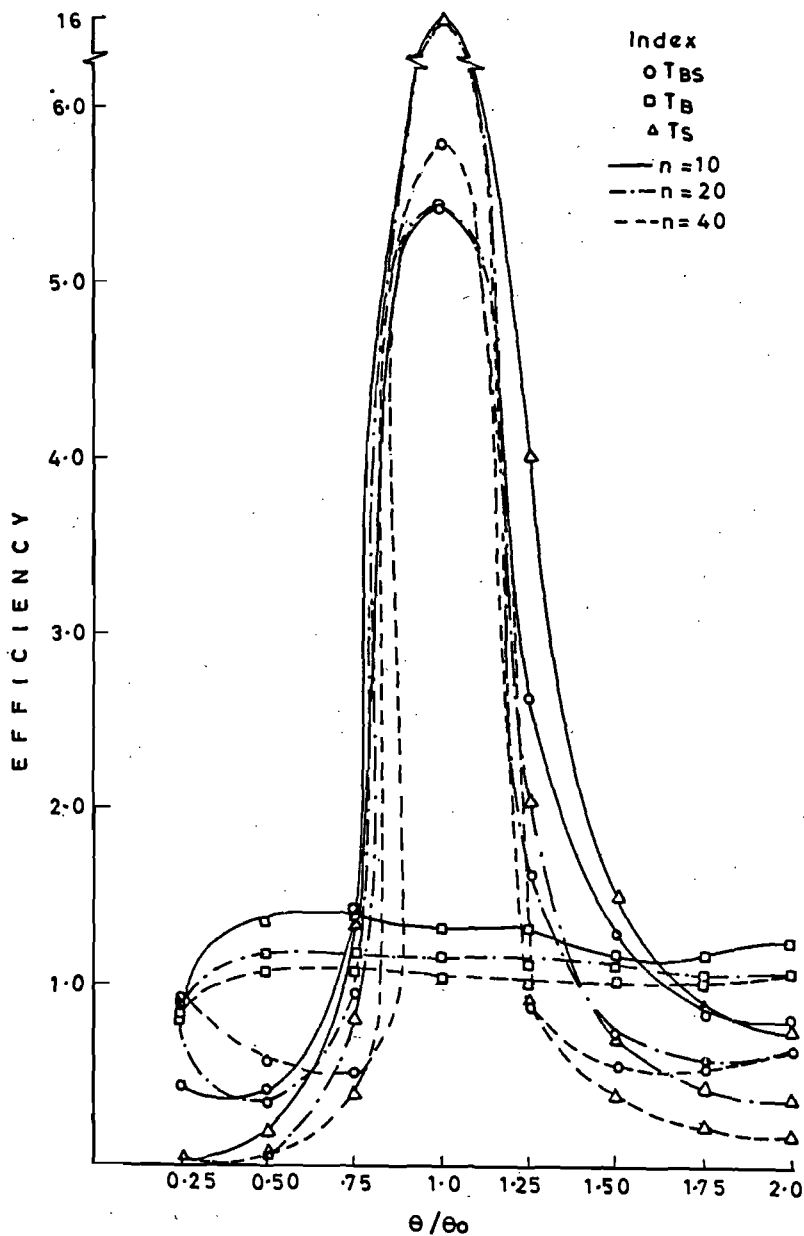


Figure 1. Efficiencies of estimators for $n=10, 20, 40$
 ($p=0.25, k=0.25, a=6, c=4$)

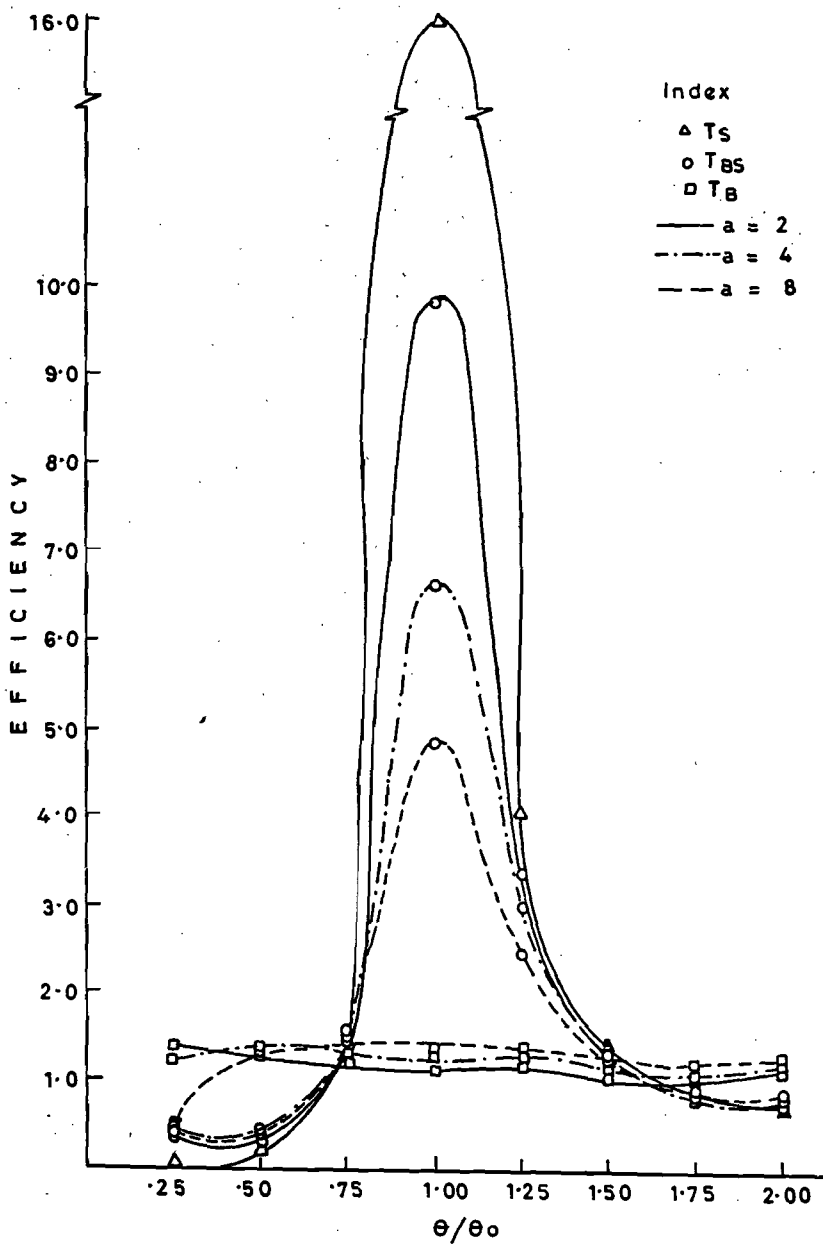


Figure 2. Efficiencies of estimators for $a=2, 4, 8$
 ($n=10, p=0.25, k=0.25, c=4$)

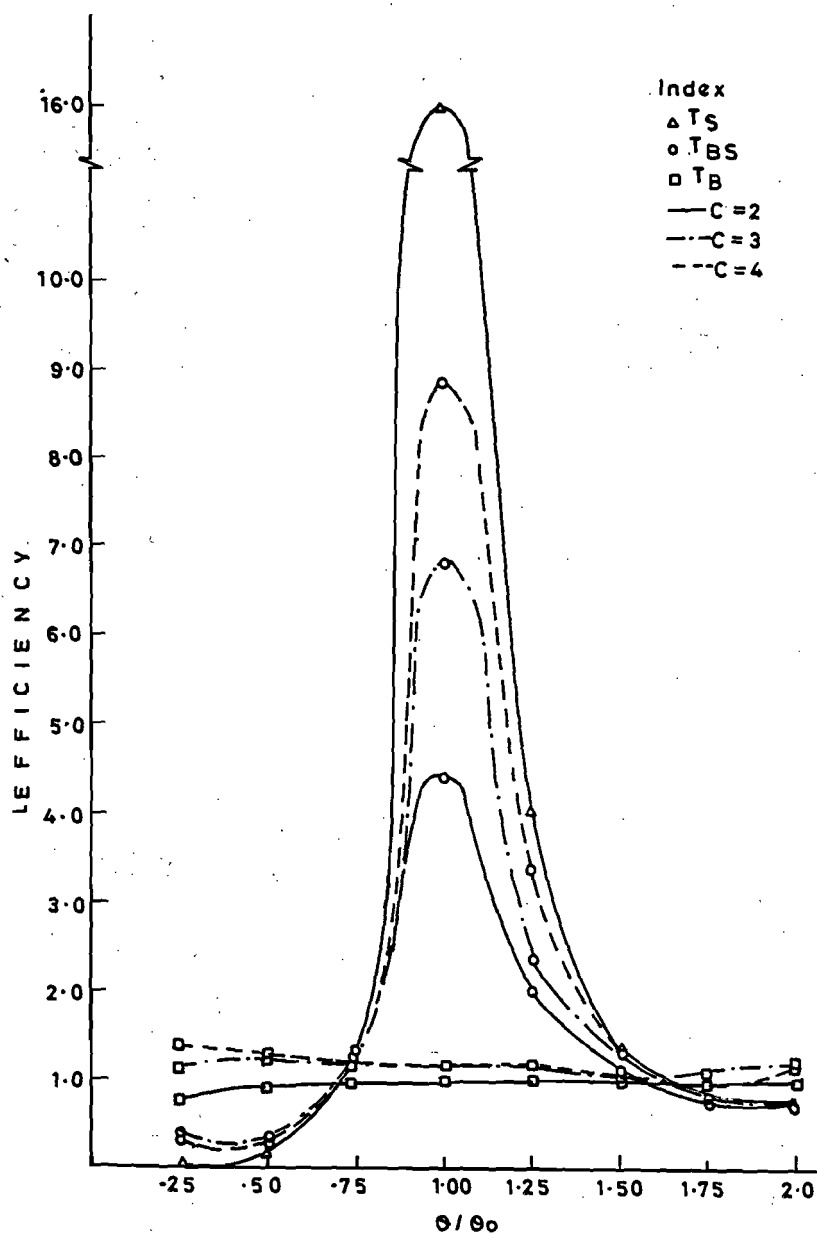


Figure 3. Efficiencies of estimators for $c=2, 3, 4$
 ($n=10, p=0.25, k=0.25, a=2$)

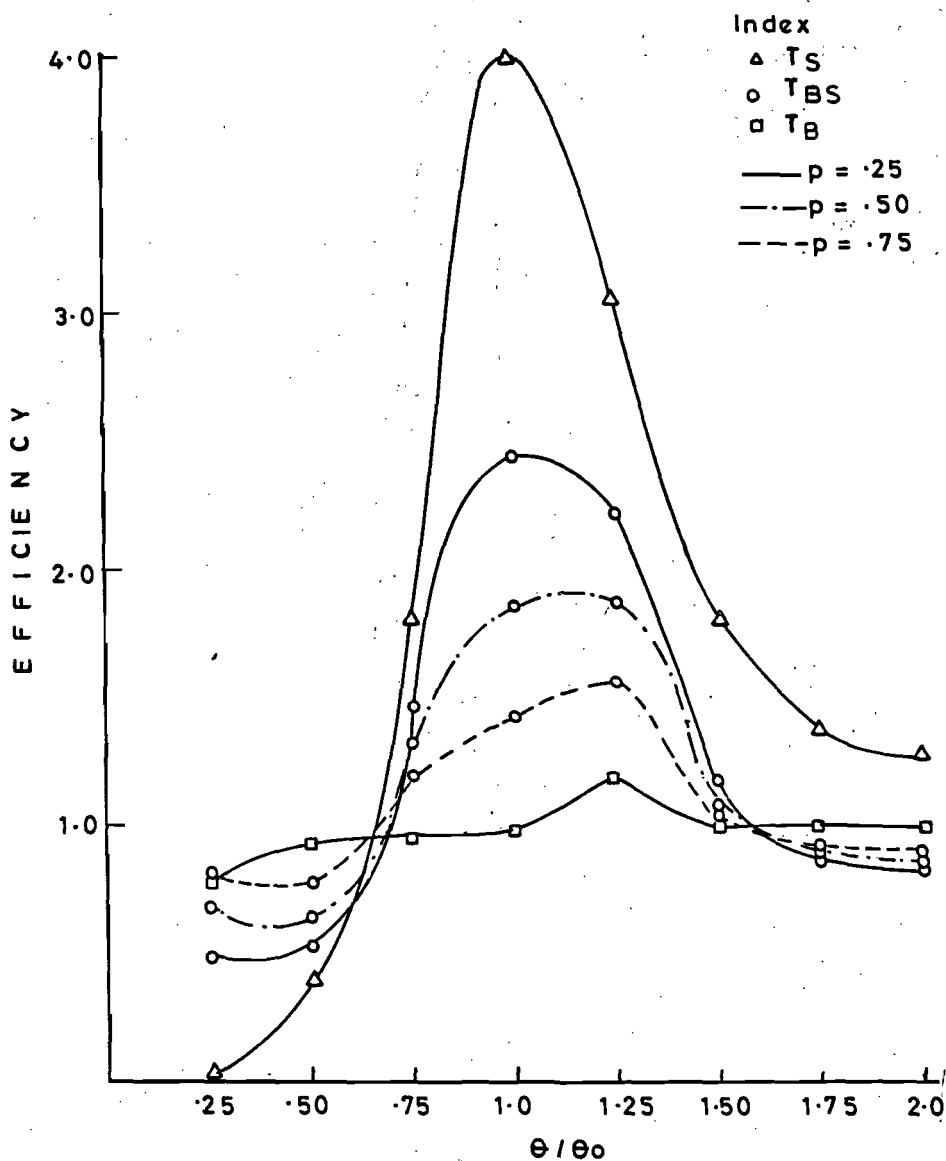


Figure 4. Efficiencies of estimators for $p=0.25, 0.50, 0.75$
 ($n=10, a=2, c=2, k=0.50$)

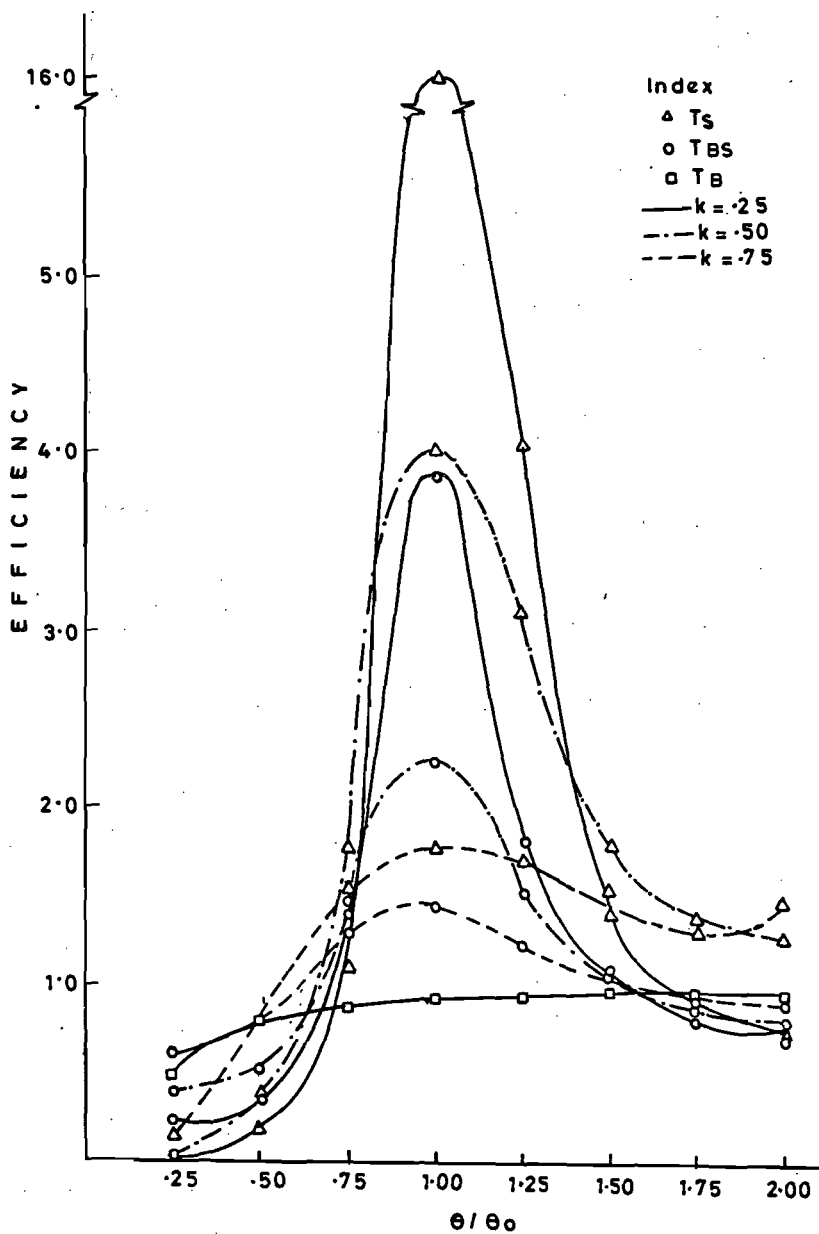


Figure 5. Efficiencies of estimators for $k=0.25, 0.50, 0.75$
 $(n=10, p=0.25, a=4, c=2)$

TABLE 1. Efficiencies of estimators T_S , T_{OB} , T_{BS} for $n = 10$, $k = 0,25$

Efficiency			θ/θ_0							
of			0,25	0,50	0,75	1,00	1,25	1,50	1,75	2,00
T_S			0,212	0,183	1,366	16,00	4,022	1,398	0,915	0,763
T_{OB}	a c									
	2 3		1,154	1,208	1,192	1,173	1,180	1,126	1,123	1,168
	2 4		1,433	1,277	1,199	1,162	1,193	1,064	1,065	0,972
	6 3		0,497	1,008	1,154	1,204	2,130	1,198	1,189	1,209
	6 4		0,801	1,413	1,426	1,361	2,669	1,221	1,198	1,274
T_{BS}	a c	p								
	2 3	0,25	0,405	0,352	1,368	6,833	2,843	1,298	0,826	0,763
	2 3	0,75	0,886	0,640	1,341	2,657	1,834	1,205	0,917	0,902
	2 4	0,25	0,325	0,298	1,328	9,868	3,424	1,376	0,855	0,746
	2 4	0,75	0,843	0,488	1,293	4,577	2,687	1,316	0,874	0,821
	6 3	0,25	0,317	0,405	1,452	4,281	2,130	1,222	0,872	0,842
	6 3	0,75	0,567	0,754	1,315	1,732	1,411	1,168	1,033	1,034
	6 4	0,25	0,450	0,432	1,446	5,433	2,669	1,320	0,887	0,848
	6 4	0,75	0,809	0,844	1,441	2,176	1,623	1,249	1,027	1,042

References

- LEMMER, H.H. (1981). From ordinary to Bayesian shrinkage estimators. *South African Statist. J.* Vol. 15, pp. 57–72.
- MEHTA, J.S. & SRINIVASAN, R. (1971). Estimation of the mean by shrinkage to a point. *J. Amer. Statist. Assoc.*, Vol. 66, pp. 86–90.
- PANDEY, B.N. (1979). On shrinkage estimation of normal population variance. *Commun. Statist.* Vol. A8, pp. 359–365.
- SINHA, S.K. & KALE, B.K. (1980). *Life testing and reliability estimation*. Wiley Eastern Limited, Delhi.
- THOMPSON, J.R. (1968). Some shrinkage techniques for estimating the mean. *J. Amer. Statist. Assoc.* Vol. 63, pp. 113–122.